

Ratan Lal Bunkar

Hardware Systems Engineer

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SUMMARY

Hardware & Systems Engineer with 3+ years in tool validation, equipment qualification, and cross-domain HW/FW debug on electromechanical platforms through NPD, NPI, and HVM readiness. Built Python-based ATE and custom fixtures on SBC platforms, cutting validation cycle time 90% and output variance 75% via DOE, SPC, and Gage R&R. M.S. EE (NTUT) and B.Tech. EE (IIT Ropar) with digital logic, VLSI, and semiconductor device coursework. Targeting Memory Test, Validation, Characterization, or Reliability Engineering.

CORE SKILLS

Test Automation & Scripting: Python (NumPy, Pandas, SciPy, Matplotlib, pytest, unittest); Raspberry Pi / SBC platforms; pyserial, bleak, pyshark; log and packet analysis; state-machine test logic

Equipment & Tool Qualification: IQ/OQ/PQ, Gage R&R, TMV, test fixture design, ATE / CATS development; bench instrumentation

Hardware, Schematic & Signal Integrity: Altium Designer (schematic capture, pin mapping, net routing); PCBA-level validation; signal-chain, power-behavior, and timing-margin analysis

Process & Data Analysis: Design of Experiments (DOE), SPC, ANOVA, capability analysis, variance reduction; Minitab (DOE / ANOVA / SPC / MSA); JMP

System-Level Debug & RCA: Cross-domain troubleshooting across hardware, firmware, and mechanical layers; structured Root Cause Analysis; failure characterization; corrective action

Requirements & Systems Engineering: Requirements elicitation and decomposition; MBSE with SysML / UML (Enterprise Architect, Cameo Systems Modeler); SRS / SDS / URS / ERS / ESRS authoring; requirements-to-test traceability

Digital & Interfaces (Coursework / Self-Learning): UART, SPI, I²C, RS-232/422/485, BLE, LTE Cat-M1, RFID; Digital logic design, Verilog; USB 2.0/3.0 signaling; Ethernet PoE; power rail / PDN design; DC-DC switching

Engineering Process & Tools: NPD→NPI, design transfer, supplier transfer, sustenance; V&V lifecycle, DFMEA, IQC / AQL sampling; PLM (Teamcenter), ALM (Polarion), JIRA, Confluence, Git, Linux

PROFESSIONAL EXPERIENCE

Hardware Systems & Test Automation Engineer — *SHL Technologies, Taoyuan, Taiwan*

May 2023 – Present

- **Owned end-to-end system definition through NPD→NPI→HVM readiness:** authored requirements, led MBSE (SysML) system design, drove full V&V; supported DFMEA, design transfer, and IQC / AQL sampling for critical components.
- **Reduced validation cycle time 90%** from connected self injection platforms by developing Python-based ATE / CATS on SBCs (Raspberry Pi) integrating sensors and BLE / UART sniffers for parametric sweeps, timing validation, and endurance runs.
- **Cut system-level performance variance 75%** via DOE-based characterization of actuator timing against spec, sweeping electrical and mechanical parameters to map operating margin and define the process window.
- **Delivered 8% cost saving** on V&V tooling spend by deploying in-house SBC-based ATE in place of commercial validation equipment.
- **Led IQ/OQ / PQ and Gage R&R** on custom test fixtures on connected health platforms, ensuring

measurement-system integrity and traceability of requirements-to-test.

- **Drove structured RCA across HW/FW/EE domains** on failure clusters (power anomalies and wireless Interfaces); built Python log and packet-capture pipelines to accelerate triage.

Research Assistant, Systems & Control Lab — *NTUT, Taipei, Taiwan*

Feb 2021 – Jan 2023

- Developed signal-processing algorithms (NMF, RPCA, REpet) in Python / PyTorch; applied FFT, filtering, and statistical validation transferable to waveform and signal-integrity analysis. Co-authored peer-reviewed paper at ICSS&E 2022; awarded Taipei Tech Scholarship and SVS Research Grant.

Control Systems Engineering Intern — *DCM Engineering Products, Punjab, India*

Mar – Aug 2019

- Programmed Siemens PLCs (Step 7, Ladder Logic) and HMI dashboards; supported commissioning, sensor wiring, signal mapping, and troubleshooting for industrial automation lines.

KEY PROJECTS (SHL Technologies)

- **SmartHub, BLE-to-Cloud System Definition & V&V:** Owned end-to-end system definition using MBSE / SysML (Cameo Systems Modeler); authored SRS, SDS, System Test Specification, BOM; led system-level integration, verification, validation, and RCA across HW/FW/cloud layers.
- **Molly cCap, BLE Timing & Latency Validation:** Designed low-cost automated test fixture measuring BLE activation time, payload accuracy, and broadcast latency; authored URS / EDS and executed IQ/OQ/PQ and Gage R&R for fixture qualification.
- **Elexy, Electromechanical Platform Characterization:** Built Power Unit Test Bench to characterize plunger force, dose accuracy, and retraction timing under variable loads; authored IQC protocols and stepper-motor validation; identified cross-domain design-phase failures.

EDUCATION

M.S. Electrical Engineering — National Taipei Univ. of Technology, Taiwan Feb 2021 – Jan 2023

Coursework: Digital Signal Processing, Advanced Control Systems, Machine Learning

B.Tech. Electrical Engineering — Indian Institute of Technology (IIT) Ropar, India Aug 2016 – Aug 2020

Coursework: Digital Electronics & Logic Design, VLSI Design, Microprocessors, Semiconductor Devices

Publication: Bunkar & Lian, “*NMF-based noise reduction over singing voices via RPCA and REpet*,” ICSS&E 2022.

Languages: English (TOEIC 825) · Hindi (Native) · Mandarin (Basic)